SET - B

12. a) Analyze why decision trees are considered interpretable compared to other models when applied to employee promotion prediction.

Dataset: Employee Promotion

Experience (years)	Performance Score	Training completed (Yes=1, No=0)	Leadership Score	Promoted (Yes=1, No=0)
3	80	1	70	1
2	65	0	60	0
5	85	1	75	1
4	78	1	72	1

→ The Interpretability of a Decision Tree stems from its ability to mimic human reasoning through a series of hierarchical logical rules.

$$\text{Entropy} = -\sum P_i \log_e P_i$$

$$\text{Information Gain} = E(S) - \sum_{v \in \{1,0\}} \frac{|S_v|}{|S|} E(S_v)$$

Step 1: Calculate the Entropy of Target Variable

Target variable = Promoted

Attributes/Features = 4

Employees = 4

Promoted (Yes = 1) : 3 Employees ; Not Promoted (No = 0): 1 Employee

Total (N) : 4

$$E(S) = -\left(\tfrac{3}{4}\log_2 \tfrac{3}{4} + \tfrac{1}{4}\log_2 \tfrac{1}{4}\right)$$

$$= -(0.75(-0.415) + 0.25(-2.0))$$

$$= \underline{\underline{0.8113}}$$

Step 2: Evaluate Entropy and Gain for Each Attribute

Experience:

✓ 3 → 1 Instance (Yes = 1, No = 0)

$E(3) = 0$

✓ 2 → 1 Instance (Yes = 0, No = 1)

$E(2) = 0$

✓ 5 → 1 Instance (Yes = 1, No = 0)

$E(5) = 0$

✓ 4 → 1 Instance (Yes = 1, No = 0)

$E(4) = 0$

$$\text{Gain}(S, \text{Experience}) = 0.8113 - (1/4(0) + 1/4(0) + 1/4(0) + 1/4(0))$$
$$= 0.8113 - 0$$
$$= 0.8113$$

Performance Score:

✓ 80: 1 Instance (Yes = 1, No = 0)

$E(80) = 0$

✓ 65: 1 Instance (Yes = 0, No = 1)

$E(65) = 0$

✓ 85: 1 Instance (Yes = 1, No = 0)

$E(85) = 0$

✓ 78: 1 Instance (Yes = 1, No = 0)

$E(78) = 0$

$$\text{Gain}(S, \text{Performance}) = 0.8113 - (1/4(0) + 1/4(0) + 1/4(0) + 1/4(0) + 1/4(0))$$
$$= 0.8113 - 0$$
$$= 0.8113$$

Training Completed:

✓ 1: 3 Instances (Yes = 3, No = 0)

$E(1) = 0$

✓ 0: 1 Instance (Yes = 0, No = 1)

$E(0) = 0$

$$Gain(S, Training) = 0.8113 - (1/4(0) + 1/4(0) + 1/4(0) + 1/4(0))$$
$$= 0.8113 - 0$$
$$= 0.8113$$

Leadership Score:

✓ 70: 1 Instance (Yes = 1, No = 0)

$E(70) = 0$

✓ 60: 1 Instance (Yes = 0, No = 1)

$E(60) = 0$

✓ 75: 1 Instance (Yes = 1, No = 0)

$E(75) = 0$

✓ 72: 1 Instance (Yes = 1, No = 0)

$E(72) = 0$

$$Gain(S, Leadership) = 0.8113 - (1/4(0) + 1/4(0) + 1/4(0) + 1/4(0))$$
$$= 0.8113 - 0$$
$$= 0.8113$$

Observations:

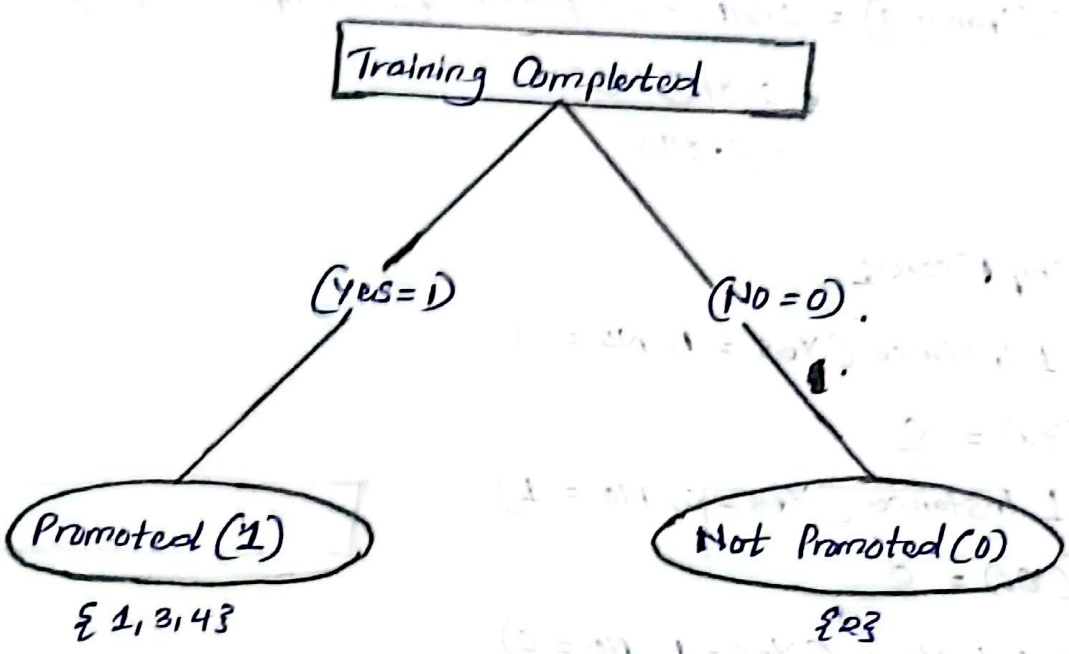
→ Each attribute perfectly splits the data.

- No further splitting is needed.
- Nodes are pure.

NIB

- If Training Completed = 1 : Employees {1, 3, 4} are all promoted (1, 1, 1).
- If Training Completed = 0 : Employee {2} is not Promoted (0)

Decision Tree



12. b)

Analyze how the independence assumption in the Naive Bayes classifier affects the accuracy of Spam email detection.

Dataset: Spam Email Detection.

Email Length (words)	Number of Links	Special Characters	Sender Reputation (Score)	Spam (Yes=1, No=0)
450	5	12	3	1
200	1	3	8	0
500	6	15	2	1
250	2	4	7	0

-> Naive Bayes classifier is a probabilistic machine learning model based on <u>Bayes Theorem.</u>

<u>Dataset:</u>

4 -> Emails, 4 -> features

<u>Let:</u>

Email Length $\Rightarrow X_1$

Number of Links $\Rightarrow X_2$

Special Characters $\Rightarrow x_3$

Sender Reputation $\Rightarrow X_4$

Target variable = Spam ($Y \in \{0, 1\}$).

Total Samples, $n = 4$

Spam emails ($Y = 1$): 2 (Emails 1 and 3).

Not Spam emails ($Y = 0$): 2 (Emails 2 and 4).

CamScanner

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

$$\hat{y} = \arg\max_{y} P(y) \prod_{i=1}^{n} P(x_i|y)$$

Prior Probability:

$P(Y=1) = 2/4 = \underline{\underline{0.5}}$

$P(Y=0) = 2/4 = \underline{\underline{0.5}}$

Feature Conditional Probabilities:

$$P(x_1, x_2, x_3, x_4 | Y) = P(x_1|Y) \cdot P(x_2|Y) \cdot P(x_3|Y) \cdot P(x_4|Y)$$

Calculate mean (μ) and variance (σ^2) for each feature:

Spam ($Y=1$):

x_1 (Length): $\mu = \frac{450+500}{2} = 475$; $\sigma^2 = \frac{(450-475)^2 + (500-475)^2}{1} = \underline{\underline{1250}}$

x_2 (Links): $\mu = \frac{5+6}{2} = 5.5$; $\sigma^2 = \frac{(5-5.5)^2 + (6-5.5)^2}{1} = \underline{\underline{0.5}}$

x_3 (Chars): $\mu = \frac{12+15}{2} = 13.5$; $\sigma^2 = \frac{(12-13.5)^2 + (15-13.5)^2}{1} = \underline{\underline{4.5}}$

x_4 (Rep): $\mu = \frac{3+2}{2} = 2.5$; $\sigma^2 = \frac{(3-2.5)^2 + (2-2.5)^2}{1} = \underline{\underline{0.5}}$

Not Spam ($Y=0$):

x_1 (Length): $\mu = \frac{200+250}{2} = 225$; $\sigma^2 = \frac{(200-225)^2 + (250-225)^2}{1} = \underline{\underline{1250}}$

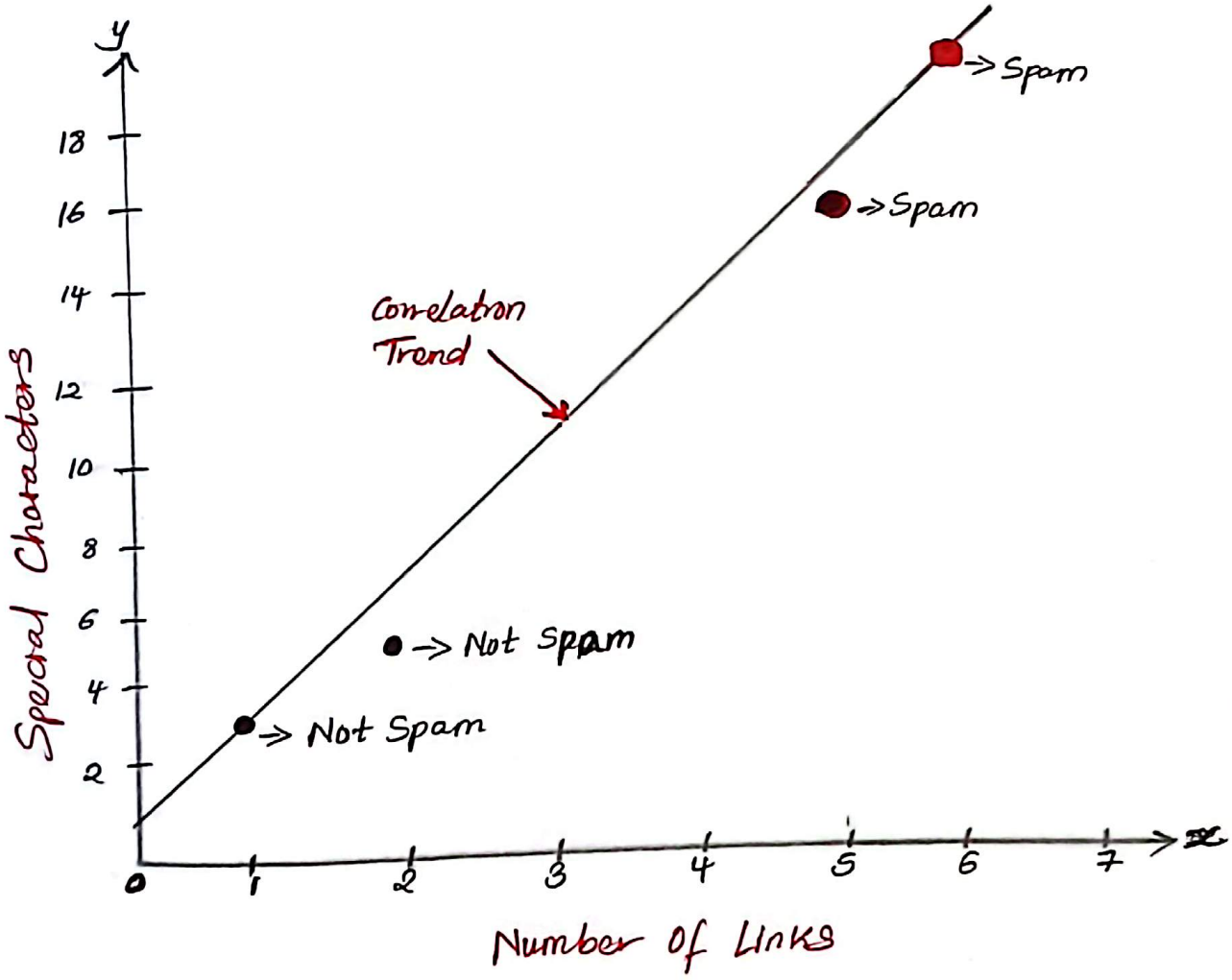
x_2 (Links): $\mu = \frac{1+2}{2} = 1.5$; $\sigma^2 = \frac{(1-1.5)^2 + (2-1.5)^2}{1} = \underline{\underline{0.5}}$

x_3 (Chars): $\mu = \frac{3+4}{2} = 3.5$; $\sigma^2 = \frac{(3-3.5)^2 + (4-3.5)^2}{1} = \underline{\underline{0.5}}$

x_4 (Rep): $\mu = \frac{8+7}{2} = 7.5$; $\sigma^2 = \frac{(8-7.5)^2 + (7-7.5)^2}{1} = \underline{\underline{0.5}}$

CamScanner

This implies that:

- The number of links (X_2) is independent of the number of special characters (X_3) once we know if the email is spam.

Relationship between Links and Special Characters in Emails.